MILITARY
SPECIFICATION

MIL-T-55031(SigC)
22 July 1958

ANTENNA GROUP AN/ARA-31 ( ) (FM HOMER); BENCH, PRE-FLIGHT AND FLIGHT TEST OF

1. SCOPE

1.1 This specification covers methods and procedures for accomplishing necessary performance and operation tests on Antenna Group AN/ARA-31 ( ) and associated components both before and after installation in an aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings of the issue in effect on date of invitation for bids form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-A-9410	Antennas, FM Communication Equipment, General Specification for Design and Placement of
MIL-R-13427	Radio Set AN/ARC-44 ( ) and Associated Components, Bench, Preflight, and Flight of
MIL-R-14304	Radio Set AN/ARC-44 and Associated Components Installation of

DRAWINGS

SIGNAL CORPS

ES-C-138998	Wiring Diagram Homing Group AN/ARA-31
ES-C-138951	Wiring Diagram AN/ARC-44 Two Station System
ES-C-138952	Wiring Diagram AN/ARC-44 Three Station System

(Copies of specifications and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

FSC 6625

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3. REQUIREMENTS

3.1 Acceptance.- Antenna Group AN/ARA-31 ( ) shall be capable of meeting the applicable tests of Section 4 both before and after installation in an aircraft. Any equipment which fails to meet tests specified herein shall be considered unacceptable for Military use.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS- The testing of equipment shall be classified as follows:

4.1.1 BENCH TESTS- Bench tests shall be conducted on each Antenna Group AN/ARA-31 ( ) at the initial installation point to determine that the complete equipment has been furnished, that the equipment has not been damaged, and that operation or performance has not been adversely affected by shipping and handling.

4.1.2 PRE-FLIGHT TEST- Preflight tests shall be conducted on each Antenna Group AN/ARA-31 ( ) at the initial installation point, after bench tests have been made, and after the equipment has been installed in the aircraft, prior to flight test. These tests shall be conducted to determine that the equipment is installed properly, to make a final check of aircraft wiring; and to check on vibration and electrical interference.

4.1.3 FLIGHT TESTS- Flight tests shall be conducted as soon as possible after pre-flight tests on all aircraft to check performance and operation of the equipment while the aircraft is in flight.

4.2 TEST EQUIPMENT REQUIRED- The following test equipment shall be used in performing the tests required by this specification:

a.27.5 Volt DCPower Supply, minimum capacity 10 amperes continuous, 25 amperes surge.

b. Radio Set AN/ARC-44 for bench testing, and installed in aircraft for preflight and flight test.

c. Panel, Signal Distribution, Radio SB-329 ( ) /AR.

d. Signal Generator SG-12 ( )/U, or equal.

e. Ground Station FM Transmitter, frequency range 24 to 51.9 mc.

f. Headset-Microphone H-101 ( )/U, or equal.

g. Headset-Cord CX-2556/U or CX-2555/U, as applicable.

h. Adopter, Coaxial Jack UG-89/U on end, Connection IPC NO. 14624 on the other end.

i. Wiring harnesses, Coaxial cables as required in accordance with typical aircraft wiring diagrams.

j. Assembly, Switch SA-474/AR,or equivalent.

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4.3 BENCH TESTS-

4.3.1 GENERAL- A test bench, cabled to accommodate the unites to be tested and various units of test equipment, shall be provided. The cabling between units shall be in accordance with a typical aircraft wiring diagram, and shall be so arranged that the component parts are accessible. The ppwer source shall be connected into the test bench wiring and shall be protected by suitable fuses or circuit breakers.

4.3.2 TEST PROCEDURES-

4.3.2.1 EXTERNAL VISUAL INSPECTION- Inspect each unit of Antenna Group AN/ARA-31( ) externally for evidence of shipping damage and that the following components are furnished. Whenever the type specification for a particular aircraft provides a list not in agreement with this list, the list in the aircraft type specifications shall govern.

Keyer KY-149/AR
Mounting MT-1620/AR
Antenna Element AT-624/AR
Network, Impedance Matching CU-459/AR
Switch Assembly SA-474/AR (where applicable).

4.3.2.2 INTERNAL VISUAL INSPECTION-

4.3.2.2.1 Remove cover of Keyer KY-149/AR. Check and inspect parts for visible damage. Interconnecting wires and cables shall be properly and securely connected. There shall be no dirt, metal scraps, or packing materials on any part of the equipment. Replace cover.

4.3.3 OPERATING TEST-

4.3.3.1 Connect Keyer KY-149/AR (with cover removed) into bench test harness. This test shall be made in conjunction with a normally operating AN/ARC-44 Radio Set, SB-329/AR, and Headset H-101 as per aircraft installation.

4.3.3.2 Set controls to the following positions.

EQUIPMENT	CONTROL	POSITION
SA-474/AR or equivalent	Homing - Comm Swith	Homing
SA-474/AR or equivalent	Squelch Disabling Sw.	Off
SB-327/ARC-44	On-Off	On
SB-327/ARC-44	Rem.-Local	Local
SB-327/ARC-44	Volume	Max.clockwise
SB-329/AR	Receiver Switches	#1 up,all others down
SB-329/AR	Channel Selector Control	any position
SB-329/AR	Volume	Normal volume control
RT-294 ( ) /ARC-44	Squelch	Adjust per MIL-R-13427

4.3.3.3 RIGHT DIPOLE KEYING- Connect the signal generator output to coaxial connector J-602, and set the generator to supply approximately 10 microvolts to the input of the keyer. Turn on the D. C. power supply. An audio tone shall be heard at a keyed rate equal to the coded letter "U" (..-).

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4.3.3.4 LEFT DIPOLE KEYING- Disconnect the signal generator from the J-602 and connect it to the J-603 coaxial connector. The audio tone shall be heard at a keyed rate equal to the coded letter "D" (-..).

4.3.3.5 ANTENNA RELAY CHECK- Disconnect the signal generator from the J-603 coaxial connector, and connect it to the J-604 coaxial connector. Place the home-communication switch in the "communication" position. Set the signal generator to supply a 1,000 cycles per second audio modulated signal of approximately 1 microvolt. The 1,000 cycles per second audio signal shall be heard in the headphones. When the homing - communication switch is placed in the homing position, the 1,000 cycles per second tone shall not be heard.

4.4 PRE-FLIGHT TESTS-

4.4.1 GENERAL- Experience has shown that ground checks are extremely unreliable with respect to accuracy of bearing indications. The following test, therefore, shall be performed solely to determine system operation. These tests shall be made with Antenna Group AN/ARA-31( )and Radio Set AN/ARC-44 ( ) installed in the aircraft as an operating system in accordance with Specifications MIL-R-13427, MIL-R-14304, and Drawings ES-C-138998 and ES-E-138951, or ES-E-138952 as applicable.

4.4.2 SET-UP PROCEDURES-

4.4.2.1 Set the controls to the following positions:

EQUIPMENT	CONTROL	POSITION
SA-474/AR or equivalent	Homing-Comm Switch	Communication
SA-474/AR or equivalent	Squelch disabling Switch	On
SB-327/ARC-44	ON-OFF	On
SB-327/ARC-44	Rem-Local	Local
SB-327/ARC-44	Volume	Max.-Clockwise
SB-329/AR	Receiver Switches	#1, up all others down.
SB-329/AR	Channel Selector Control	#1 (FM)
SB-329/AR	Volume	Normal Volume Level.
RT-294/ARC-44	Squelch	Adjust per MIL-R-13427

4.4.2.2 Start the aircraft engine (s) and operate at RPM sufficient to cause the generator (s) to charge the batteries, preferably at cruising speed. Turn the interphone "ON-OFF" switch to the "ON" position. Turn on all other electronic equipment to be operated in the aircraft. This test shall be performed after the preflight test of the Radio Set AN/ARC-44 is completed and the squelch controls are properly set.

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4.4.2.3 With the aircraft located 200 feet or more from the transmitter, adjust Radio Set AN/ARC-44 ( ) to the test frequency. Request an unmodulated carrier on this frequency, and place the "homing-communication" switch in the "homing" position. Determine that the 400 cycle tone signal is present in the headset, and that the keyer is operational. This can be done by using the main landing gear as the pivot point and rotating the aircraft thru a 380 degree circle. At some compass point, or points, the coded letter "D" (-..) the coded letter "U" (..-) and a steady 400 cycle tone signal should be received. Do not check for accuracy on the ground as reflections and proximity of ground affect antenna patterns and make accurate ground checks unreliable.

4.5 FLIGHT TESTS- Flight tests on all aircraft shall consist of normal operational use of Antenna Group AN/ARA-31 ( ) during flight tests of the aircraft.

4.5.1 METHOD OF TESTS-

4.5.1.1 SEQUENCE- After the equipment has successfully completed the requirements of the bench and preflight tests, the following flight tests shall be conducted. Flight tests shall be conducted on each equipment installed in an aircraft.

4.5.1.2 ACCEPTANCE OF FLIGHT TESTS- The equipment shall be flight tested to insure satisfactory performance and operation while the aircraft is in flight. This test shall be conducted to simulate ultimate operational environment and to check on vibration and electrical interference. This test shall be performed on all aircraft.

4.5.2 PROTOTYPE PERFORMANCE TEST PROCEDURE-

4.5.2.1 HOMING FUNCTION TEST- Advise the ground station that the "Homing Test" is to be conducted. Request a continuous keyed unmodulated signal from the ground transmitter using an authorized frequency. When on "Homing Function" the aircraft FM transmitter is inoperative, and the received signal is affected by the keyer coding, and the 400 cycles per second modulation tone. Communication shall be maintained with the "Homing" station via the UHF or VHF equipment in the aircraft.

4.5.2.2 Place the "Communicator-Homing" switch in the "Homing" position.

4.5.2.3 The coded letter "D" (-..) shall be received when the ground test station is to the left of the aircraft.

4.5.2.4 The coded letter "U" (..-) shall be received when the ground test station is to the right of the aircraft.

4.5.2.5 A steady 400 cycles per second tone signal shall be received when the aircraft on course or on the reciprocal course.

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4.5.2.6 The on-course signal, or straight line (underno wind condition) to the station shall be obtained by turning the aircraft in the direction indicated by the received coded letter ("D" left, "U" right). The "True Homing Course" shall have an aircraft bearing of zero±10 degrees. The "Reciprocal Homing Course" shall have a bearing of 180 degrees±10 degrees from the "True Homing Course."

4.5.2.7 The code characters (D and U) shall be clearly distinguishable throughout the 150 degrees sectors between the homing and the reciprocal homing courses on the respective sides of the aircraft. These sectors shall be free of false homing course indications except where shown to be caused by banking or other transient attitudes of the aircraft. Lack of symetry can be caused by autenna cables of different lengths, and also by faulty connections or faulty dipole assembly. These defects can cause various off-course indications due to unequal distribution in the homing system.

4.5.2.8 Flight tests shall be made at three altitudes specified in Specification MIL-A-9410, preferably the low, medium and high altitudes listed therein. Satisfactory homing performance shall be obtained at ranges not less than 85 percent of the corresponding ranges specified for FM communication tests. Performance shall be tested at 3 FM frequencies, preferably at low, middle, and high areas in the band.

4.5.2.9 In helicopters, rotor modulation shall be checked to determine bearing interference in accordance with Specification MIL-A-9410. It will be acceptable for the above tests to be conducted by careful aural observation.

4.5.3 PRODUCTION ACCEPTANCE FLIGHT TEST PROCEDURES-

4.5.3.1 Acceptance flight test shall consist of those tests in 4.5.2.1 through 4.5.2.5. Satisfactory results on one frequency at a distance of approximately 10 miles shall be considered sufficient for the purposes of the acceptance flight test.

4.6 FINAL INSPECTION- After conclusion of the flight tests, a careful inspection of all component parts of the installation shall be made, with particular emphasis on the cables, shock mountings, and securing brackets. All electrical connections shall be checked to make certain they are tight. Any mounting or connections which may have become loosened by vibration in flight shall be properly secured. The shock mountings and cables shall be checked for freedom of movement to make sure each unit of the system has maintained its shock isolation. No equipment need by removed for this inspection unless a mechanical or electrical failure has occurred, or is suspected to have occurred during the flight test.

4.7 ACCEPTANCE-

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4.7.1 GENERAL- Upon completion of all the aforementioned tests, and upon assurance that all units have performed satisfactorily, the equipment covered herein shall be considered acceptable.

4.7.1.1 Should the homing system fail to function properly, a check shall be made to determine that all cables are properly connected and the switches and circuit breakers are in correct position. If this does not correct the trouble and it appears to be faulty equipment, the equipment shall be returned for bench test.

5. N/A

6. N/A

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